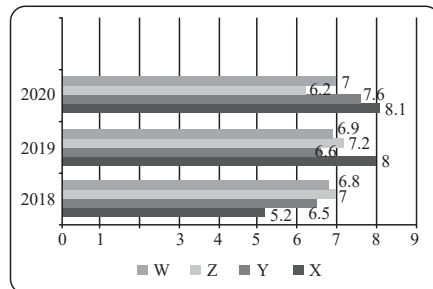


NUMERICAL AND MATHEMATICAL ABILITY

- The average monthly expenditure of a family was ₹8,600 during the first three months, ₹9,750 during the next four months and ₹9,920 during the last five months of a year. If the total savings during the year was ₹65,300, the average monthly income (in ₹) of the family was:
(a) 14,975 (b) 15,794
(c) 15,479 (d) 14,597
- A thief saw a policeman, and he started running at a speed of 18 m/s. After 12 seconds, the policeman starts running behind the thief at a certain speed. If the policeman covered 864 meters distance to reach the thief, then the speed (in m/s) of the policeman is:
(a) 28 (b) 15 (c) 30 (d) 24
- The ratio of the present age of A and B is 3 : 1. Five years ago, the ratio of their ages was 5 : 1. The age of A (in years) after eight years will be:
(a) 30 (b) 22 (c) 38 (d) 40
- Two trains, 148 m and 152 m long, are running towards each other on parallel lines at 62 km/h and 46 km/h, respectively. In what time will they be clear of each other from the moment they meet?
(a) 5 sec (b) 20 sec
(c) 10 sec (d) 15 sec
- The area of an equilateral triangle is $\frac{\sqrt{5043}}{4}$ sq. cm. The side of the equilateral triangle is :
(a) $\sqrt{23}$ cm (b) $\sqrt{29}$ cm
(c) $\sqrt{47}$ cm (d) $\sqrt{41}$ cm
- The average monthly income of A and B is ₹3,590. The average monthly income of B and C is ₹4,250. The average monthly income of A and C is ₹6,500. What is the monthly income of C?
(a) ₹7,545 (b) ₹6,984
(c) ₹8,190 (d) ₹7,160
- In a 100-marks test, the average of marks scored by 50 students was found to be 65. Later on, it was discovered that a score of 88 was misread as 68. Find the correct average.
(a) 65.60 (b) 65.50
(c) 65.70 (d) 65.40
- A man sells an article at a certain price and suffers a loss of 20%. If he sells the article at ₹100 more, he gets a profit of 30%. Find the cost price of the article (in ₹).
(a) 250 (b) 300 (c) 400 (d) 200
- A sum invested at 5% simple interest per annum grows to ₹806 in 6 years. If the amount remains the same, how much will it grow at a 12% simple interest rate in 5 years?
(a) ₹992 (b) ₹985
(c) ₹1,000 (d) ₹1,120
- Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 7% per year rate of interest for 6 years.
(a) 1430 (b) 1470
(c) 1510 (d) 1410
- The area of a regular hexagon is $2048\sqrt{3}$ cm². What is the length (in cm) of each side of the hexagon ?
(a) $32\sqrt{3}$ (b) $\frac{64\sqrt{3}}{3}$
(c) $\frac{32\sqrt{3}}{3}$ (d) $64\sqrt{3}$
- Rice worth ₹43/kg and ₹67/kg are mixed with a third variety in the ratio 2 : 1 : 5. If the mixture is worth ₹96/kg, the price (in ₹) of the third variety of rice per kg will be:
(a) 131 (b) 159 (c) 123 (d) 142
- Three numbers are in the ratio of 6 : 7 : 3 and their LCM is 3654. Their HCF is:
(a) 28 (b) 87 (c) 29 (d) 27
- Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 7% per year rate of interest for 5 years.
(a) 1050 (b) 990
(c) 1010 (d) 1090
- 20 identical solid hemispheres of radius 3 cm are melted to form a big solid hemisphere. What is the radius (in cm) of the biggest hemisphere formed, if 20% of the solid is wasted during the process?
(a) $3\sqrt{2}$ (b) 6
(c) $6\sqrt{2}$ (d) 3
- The following bar graph shows the production of cement (in 10,000 tones) by

4 companies X, Y, Z and W during the period 2018 - 2020. Study the bar graph carefully and answer the question.

Which company has the maximum average production over the course of 3 years?



(a) Y (b) X (c) W (d) Z

- Riya offers 12% discount on all her goods and offers an additional discount of 6% to those customers who pay in cash. How much will a customer have to pay for a fan marked for ₹2,400 if he pays in cash?
(a) ₹1,985.28 (b) ₹1,900
(c) ₹2,300.50 (d) ₹2,122.50
- In a college, 20% of the number of boys are equal to the three-fourth of the number of girls. What is the ratio of boys and girls in that college ?
(a) 15 : 4 (b) 13 : 4
(c) 5 : 3 (d) 7 : 6
- Atul sold two sacks of rice for ₹19,200 each. On one, he gained 20% and on the other, he lost 20%. Find his gain or loss percentage in the whole transaction.
(a) 1.8% profit (b) 7% profit
(c) 4% loss (d) 5% loss
- Study the given table and answer the question that follows.

The given table shows the number of new employees added to different categories of employees in a company for four years and the number of employees from these categories who left the company every year.

Year	Managers		Technicians		Accountants	
	New	Left	New	Left	New	Left
2001	280	120	260	50	200	100
2002	300	92	340	160	228	95
2003	310	150	450	200	254	125
2004	420	120	480	180	260	120

During the period between 2001 and 2004, the total number of technicians who left the company is what percentage (rounded off to the nearest integers) of the total number of technicians who joined the company?

(a) 39% (b) 35% (c) 24% (d) 29%

REASONING ABILITY AND PROBLEM SOLVING

21. What will come in the place of '?' in the following equation, if '+' and '-' are interchanged and '×' and '÷' are interchanged?

$$39 \div 3 - 24 + 60 \times 12 = ?$$

- (a) 163 (b) 86
(c) 136 (d) 190

22. In a certain code language, 'I swim fast' is coded as 'cp lk bs' and 'may I go' is coded as 'ap sb lk'. How is 'I' coded in that code language?

- (a) bs (b) lk
(c) sb (d) cp

23. Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). A is sitting second to the right of E. Only one person is sitting between D and A. B is sitting third to the right of G. F is an immediate neighbour of B. How many people are sitting between D and E when counted from the left of E?

- (a) Two (b) One
(c) Four (d) Three

24. Select the correct mirror image of the given figure when the mirror is placed at MN as shown below.



- (a) a I 0 Я H (b) s I 0 Я H
(c) 1 a 0 Я H (d) I 9 0 Я H

25. In a certain code language, 'WING' is coded as '2618' and 'NEAT' is coded as '3469'. What is the code for 'N' in that language?

- (a) 6 (b) 8 (c) 1 (d) 3

26. Select the set in which the numbers are related in the same way as the numbers of the following sets.

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13, can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

- (4, 35, 11) (1, 82, 61)
(a) (13, 53, 20) (b) (8, 33, 15)
(c) (2, 53, 6) (d) (21, 310, 23)

27. In a code language, 'CAKE' is written as 'ACIG' and 'GOLD' is written as 'EQJF'. How will 'WIRE' be written in that language?

- (a) GUPK (b) UKPG
(c) GUKP (d) GPUK

28. Select the figure from among the given options that can replace the question mark (?) in the following series and complete the pattern.

A C K	C K 8	Δ P K	P K 8	
P ↑ 8	Δ P Z	Δ Z 8	↑ Δ C	
Δ Δ Z	Δ ↑ A	↑ A C	A Z Δ	?

- (a)

↑ 8 C
A K Δ
Z Δ P

 (b)

↑ P K
A Z Δ
8 Δ C

- (c)

↑ Δ K
A C 8
Z Δ P

 (d)

Δ Δ K
P C 8
↑ A Z

29. Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements.

Statements:

Some chickens are hens.

No hen is an egg.

Conclusions:

- I. No chicken is an egg.
II. All hens are chickens.
(a) Only conclusion II follows
(b) Both conclusions I and II follow
(c) Neither conclusion I nor II follows
(d) Only conclusion I follows

30. Identify the figure from the given options that when put in place of the question mark (?) will logically complete the series.

∇	Φ	☆	∇	Φ	☆	∇	Φ	☆	?
∇	☆	Φ	∇	☆	Φ	∇	☆	Φ	

- (a)

∇	Φ
☆	∇

 (b)

☆	∇
Φ	∇

- (c)

☆	∇
Φ	∇

 (d)

Φ	∇
☆	∇

31. Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the center (but not necessarily in the same order). C is sitting to the immediate right of E. D is sitting third to the left of C. A is the immediate neighbour of B and C. F is not the immediate neighbour of E. What is the position of G with respect to A?

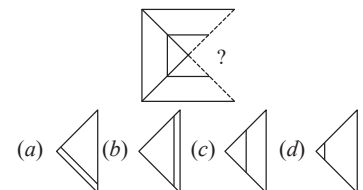
- (a) Second to the left
(b) Third to the left
(c) Second to the right
(d) To the immediate left

32. 7 is related to 30 following a certain logic. Following the same logic, 14 is related to 100. To which of the following is 22 related following the same logic?

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

- (a) 189 (b) 190 (c) 180 (d) 170

33. Select the option figure that will replace the question mark (?) in the figure given below to complete the pattern.



34. What should come in place of '?' in the given series?

$$-12, +24, -48, +96, ?, +384$$

- (a) -192 (b) 108 (c) 96 (d) -108

35. What will come in place of the question mark (?) in the following equation if '+' and '-' are interchanged and '×' and '÷' are interchanged?

$$5 \div 6 \times 3 - 4 + 11 = ?$$

- (a) 7 (b) 11 (c) 14 (d) 3

36. JM12 is related to EH15 in a certain way. In the same way, QT34 is related to LO37. To which of the following is SV23 related following the same logic?

- (a) MP26 (b) MQ27
(c) NQ26 (d) NP27

37. Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s).

Statements:

No bus is a car.

No car is a truck.

Conclusions:

- I. No car is a bus.
II. No truck is a car.
(a) Only conclusion II follows.
(b) Only conclusion I follows.
(c) Neither conclusion I nor II follows.
(d) Both conclusions I and II follow.

38. In a code language, 'COLONY' is coded as '48' and 'INTERCHANGE' is coded as '88'. How will 'MOUNTAIN' be coded in the same language?
(a) 64 (b) 86 (c) 66 (d) 84
39. HFLI is related to OMSP in a certain way based on the English alphabetical order. In the same way, KIOL is related to RPVS. To which of the following is MKQN related, following the same logic?
(a) TRUX (b) RTXU
(c) RTUX (d) TRXU
40. What should come in place of ? in the given series based on the English alphabetical order?
ACE, FHJ, KMO, PRT, ?
(a) NOP (b) UWY
(c) GJK (d) BCD

GENERAL AWARENESS

41. The Pharmacy (Amendment) Bill, 2023 was introduced in Lok Sabha on 3 August 2023, which amends the _____.
(a) Pharmacy Act, 1950
(b) Pharmacy Act, 1949
(c) Pharmacy Act, 1947
(d) Pharmacy Act, 1948
42. In which state/union territory is Losar celebrated to mark the start of the new year in that region?
(a) Kerala (b) Bihar
(c) Ladakh (d) New Delhi
43. Kanak Rele, who established Nalanda Dance Research Centre, was a performer of which of the following classical dance forms of India?
(a) Mohiniyattam (b) Manipuri
(c) Kathak (d) Sattriya
44. Which of the following acts introduced the system of dyarchy in the provincial administration in India?
(a) The Government of India Act of 1909
(b) The Government of India Act of 1919
(c) The Government of India Act of 1935
(d) The Government of India Act of 1947
45. In which part of India is cool mountain climate observed?
(a) West (b) South
(c) North (d) East
46. Who was the founder of the Mauryan dynasty?
(a) Vikramaditya
(b) Bindusara
(c) Chandragupta Maurya
(d) Ashoka
47. As of 31st October 2023, the senior-most member (in terms of age) in the 17th Lok Sabha, Shafiqur Rahman Barq represents which political party?
(a) Bharatiya Janata Party
(b) Indian National Congress
(c) Samajwadi Party
(d) Rashtriya Janata Dal
48. Which of the following books is written by Dadabhai Naoroji?
(a) Poverty and India
(b) Important Chapters of Poverty
(c) Poverty and Un-British Rule in India
(d) Economics History of India
49. The historical fiction 'The Glory of Patan' by KM Munshi was originally written in which Indian language?
(a) Malayalam (b) Bengali
(c) Tamil (d) Gujarati
50. On which date did Punjab abolish the Legislative Council?
(a) 1 January 1971 (b) 1 January 1977
(c) 1 January 1975 (d) 1 January 1970
51. The chemical compound present in turmeric is:
(a) curcumin (b) tartaric acid
(c) carotene (d) capsaicin
52. Who among the following established the Indore gharana?
(a) Bhimsen Joshi
(b) Ustad Amir Khan
(c) Tansen
(d) Ustad Ghaghe Khuda Baksh
53. With reference to medieval Indian architecture style, what is 'Pietra Dura'?
(a) Sculpture art of the Mauryan period
(b) Oil paintings on the walls of Taj Mahal
(c) Fine quality clothes used by the Mughals
(d) Decorating the walls with floral designs made of semi-precious stones
54. The Rabi season of agriculture begins in October-November and ends in _____.
(a) June-July
(b) March-April
(c) January-February
(d) May-June
55. What is the chemical formula for sodium bicarbonate?
(a) NaCl (b) Na₂CO₃
(c) NaHCO₃ (d) NaOH
56. Which of the following is one of the features of capital goods?
(a) They are durable.
(b) They are inexpensive.
(c) They are Giffen goods.
(d) They are inferior goods.
57. Which is the correct descending order of the Padma Awards?
(a) Padma Shri, Padma Vibhushan, Padma Bhushan
(b) Padma Bhushan, Padma Shri, Padma Vibhushan

- (c) Padma Vibhushan, Padma Bhushan, Padma Shri
(d) Padma Shri, Padma Bhushan, Padma Vibhushan
58. Which of the following dynasties established the Vijayanagara kingdom in fourteenth century?
(a) Aravidu dynasty
(b) Saluva dynasty
(c) Tuluva dynasty
(d) Sangama dynasty
59. The red colour of tomatoes is due to the presence of:
(a) limonene (b) lycopene
(c) beta carotene (d) alizarin
60. Which Union Territory had minimum literacy rate as per 2011 Census?
(a) Chandigarh
(b) Dadra and Nagar Haveli
(c) Lakshadweep
(d) Delhi
61. In 1976, in which Part of the Indian Constitution under Article 51A were Fundamental Duties added?
(a) VI (b) V (c) IV A (d) III
62. With which sport is Sanjana Bathula associated?
(a) Roller skating (b) Squash
(c) Shooting (d) Badminton
63. Who was the captain of Royal Challengers Bangalore in the Women's Premier League event 2023?
(a) Harmanpreet Kaur
(b) Deepti Sharma
(c) Shafali Verma
(d) Smriti Mandhana
64. Which of the following statements is/are correct about the famous place Mawsynram?
I. It is the wettest place on the earth.
II. It is located in Meghalaya state of India.
(a) Only I is correct.
(b) Neither I nor II is correct.
(c) Only II is correct.
(d) I and II both are correct.
65. How many countries share their land borders with India?
(a) Nine (b) Seven
(c) Six (d) Eight

ENGLISH LANGUAGE AND COMPREHENSION

66. Select the INCORRECTLY spelt word.
(a) Trifle (b) Surrender
(c) Consicstent (d) Disgust
67. Select the most appropriate synonym of the given word.
Ecstatic
(a) Shocked (b) Happy
(c) Angry (d) Interested

68. The following sentence has been split into four segments. Identify the segment that contains a phrasal verb error.
Mrs. Premalatha asked / Malini to hand down / all the / payment slips.
(a) all the
(b) Mrs. Premalatha asked
(c) Malini to hand down
(d) payment slips
69. What does it imply when people say to 'read between the lines'?
(a) To read something fast
(b) To read with an ability to infer the meaning
(c) To skip initial and final lines in a paragraph
(d) To avoid detailed information
70. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.
The workers had been repairing / the mall road in / town for / several days now.
(a) the mall road in
(b) several days now
(c) The workers had been repairing
(d) town for
71. Select the most appropriate meaning of the given idiom.
A piece of cake
(a) A slice of cake for eating
(b) Easy to do or achieve
(c) A cake which is sold as a piece
(d) A cake that broke into pieces
72. Select the most appropriate option to fill in the blank.
Cultural heritage is the lifeblood of any _____ and serves as a link between generations.
(a) lethargy (b) birthright
(c) civilisation (d) believe
73. Select the most appropriate verb form to fill in the blank.
Arun _____ his bicycle to school every day.
(a) ride (b) riding
(c) rides (d) rode
74. Select the most appropriate synonym of the word 'pursue' from the given sentence.
Dr. Kalam's success in the field of science has inspired young Indians to dream big and chase those dreams with immense hard work.
(a) chase (b) inspired
(c) immense (d) success
75. The following sentence has been split into four segments. Identify the segment that contains a grammatical error. My secretary is disposed of / discuss everything in great detail / before she begins / the actual work.
(a) My secretary is disposed of
(b) discuss everything in great detail
(c) before she begins
(d) the actual work.
76. Select the option that correctly rectifies the underlined spelling error.
The accideent on the highway caused a long traffic jam.
(a) accidnt (b) acident
(c) accident (d) acident
77. The following sentence has been divided into four segments. One of them contains an error. Select the segment that contains the error from the given options.
Neither Smith / nor Rose / were in / the classroom.
(a) Neither Smith (b) were in
(c) the classroom (d) nor Rose
78. Select the most appropriate synonym of the underlined word
The project's tight deadline and complex requirements made it a gruelling task for the team.
(a) Empowering (b) Inclusive
(c) Challenging (d) Accessible
79. Select the most appropriate option to fill in the blank.
He is _____ tired to finish his homework.
(a) to (b) too
(c) most (d) more
80. Select the most appropriate option to fill in the blank.
Please listen to all announcements _____.
(a) easily (b) generally
(c) hungrily (d) attentively
81. Select the most appropriate synonym of the given word.
Energy
(a) Frailty (b) strife
(c) Speed (d) Power
82. Select the most appropriate synonym of the underlined word.
The detective's job was to discern the truth from the web of lies.
(a) Ignore (b) Confuse
(c) Perceive (d) Fabricate
83. Select the sentence that has a grammatical error.
(a) She always eats healthy meals.
(b) The sun rises in the east.
(c) We watches movies every Friday night.
(d) He works as a software developer.
84. Select the option that correctly rectifies the underlined spelling error.
The rainbow only appears after a hurican.
(a) hurrecane (b) hurrecain
(c) hurrican (d) hurricane
85. Select the most appropriate option to fill in the blank.
Accepting how wonderful you truly are and incorporating this _____ into your attitude and personality is perhaps one of the most difficult things to do in life.
(a) insight (b) craziness
(c) folly (d) glitch

Directions (86-90): In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank.

My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to _____ 1 _____. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come _____ 2 _____ walks with us. This, however, we had to stop for, _____ 3 _____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4 _____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____ 5 _____.

86. Select the most appropriate option to fill in blank number 1.
(a) fly (b) read
(c) flea (d) file
87. Select the most appropriate option to fill in blank number 2.
(a) by (b) on
(c) at (d) in
88. Select the most appropriate option to fill in blank number 3.
(a) neither (b) either
(c) never (d) ever
89. Select the most appropriate option to fill in blank number 4.
(a) piece (b) peace
(c) pace (d) space
90. Select the most appropriate option to fill in blank number 5.
(a) indignation (b) intimation
(c) inclination (d) ignition



Explanations

1. (a) Sum of expenditure of first three months
 $= 8600 \times 3 = ₹25800$
 Sum of next 4 months $= 9750 \times 4 = ₹39000$
 Sum of last 5 months $= 9920 \times 5 = ₹49600$
 Total expenditure $= 25800 + 39000 + 49600$
 $= ₹1,14,400$
 Total income $= 1,14,400 + 65300$
 $= ₹1,79,700$

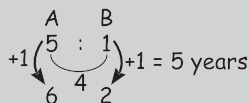
$$\therefore \text{Required average} = \frac{179700}{12} = ₹14975$$

2. (d) Distance covered by thief in 12 sec
 $= 18 \times 12 = 216 \text{ m}$
 Time taken by policeman to catch the thief
 $= \frac{864 - 216}{18} = 36 \text{ sec}$

$$\text{Speed of policeman} = \frac{864}{36} = 24 \text{ m/sec}$$

3. (c) Let present age of A $= 3x$ years
 Present age of B $= x$ years
 According to question,
 $\frac{3x-5}{x-5} = \frac{5}{1}$
 $\Rightarrow 3x - 5 = 5x - 25 \Rightarrow x = 10$
 Age of A after 8 years $= 10 \times 3 + 8 = 38$ years

TOPPER'S GUIDANCE



Age of A after 8 years $= 6 \times 5 + 8$
 $= 38$ years

4. (c) Total distance $= 148 + 152 = 300 \text{ m}$
 Total speed $= 62 + 46 = 108 \text{ km/h}$
 $= 108 \times \frac{5}{18} = 30 \text{ m/sec}$
 $\therefore$ Required time $= \frac{300}{30} = 10 \text{ sec}$
5. (d) Area of equilateral triangle $= \frac{\sqrt{3}}{4} \times \text{side}^2$
 $\Rightarrow \frac{\sqrt{3}}{4} \times \text{side}^2 = \frac{\sqrt{5043}}{4}$
 $\Rightarrow \text{Side}^2 = \frac{\sqrt{5043}}{\sqrt{3}} = \sqrt{1681}$
 $\Rightarrow \text{Side}^2 = 41 \Rightarrow \text{Side} = \sqrt{41} \text{ cm}$
6. (d) Sum of A and B $= 3590 \times 2 = ₹7180$
 Sum of B and C $= 4250 \times 2 = ₹8500$
 Sum of A and C $= 6500 \times 2 = ₹13000$
 Sum of (A + B + C) $= \frac{7180 + 8500 + 13000}{2}$

$$= \frac{28680}{2} = ₹14340$$

$$\text{Monthly income of C} = 14340 - 7180 = ₹7160$$

7. (d) Sum of marks $= 65 \times 50 = 3250$
 $\therefore$ Correct average $= \frac{3250 + 88 - 68}{50}$
 $= \frac{3270}{50} = 65.4$

TOPPER'S GUIDANCE

$$\begin{aligned} \text{Correct average} &= 65 + \frac{88 - 68}{50} \\ &= 65 + \frac{2}{5} = 65.4 \end{aligned}$$

8. (d) Let cost price of an article $= ₹x$
 According to question,
 $(100 + 30)\% \text{ of } x - (100 - 20)\% \text{ of } x = 100$
 $\Rightarrow 130\% \text{ of } x - 80\% \text{ of } x = 100$
 $\Rightarrow 50\% \text{ of } x = 100$
 $\Rightarrow x \times \frac{1}{2} = 100 \Rightarrow x = 200$

TOPPER'S GUIDANCE

If a loss of $a\%$ turns into profit of $b\%$ by increasing the selling price of an article by $₹x$, then

$$\begin{aligned} \text{Cost price of an article} &= \frac{x}{b+a} \times 100 \\ \therefore \text{Cost price} &= \frac{100}{30+20} \times 100 = ₹200 \end{aligned}$$

9. (a) Let Principal be $₹P$.
 According to question,
 $P + \frac{P \times 5 \times 6}{100} = 806$
 $\Rightarrow 100P + 30P = 806 \times 100$
 $\Rightarrow P = \frac{80600}{130} = ₹620$
 Now, Amount $= 620 + \frac{620 \times 12 \times 5}{100}$
 $= 620 + 372 = ₹992$

TOPPER'S GUIDANCE

$$\begin{aligned} (100 + 6 \times 5)\% \text{ of } P &= ₹806 \\ \therefore (100 + 12 \times 5)\% \text{ of } P &= \frac{806}{130} \times 160 \\ &= ₹992 \end{aligned}$$

10. (b) According to question,
 S.I. $= \frac{3500 \times 7 \times 6}{100} = ₹1470$

11. (b) Let side of hexagon $= x \text{ cm}$

$$\text{Area of hexagon} = 6 \times \frac{\sqrt{3}}{4} x^2$$

$$\Rightarrow 2048\sqrt{3} = 6 \times \frac{\sqrt{3}}{4} x^2$$

$$\Rightarrow x^2 = \frac{4096}{3} \Rightarrow x = \frac{64}{\sqrt{3}}$$

$$\Rightarrow x = \frac{64}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{64\sqrt{3}}{3}$$

12. (c) Total quantity of mixture $= (2 + 1 + 5)$
 $= 8 \text{ kg}$
 According to question,
 $43 \times 2 + 67 \times 1 + 5 \times x = 96 \times 8$
 $\Rightarrow 86 + 67 + 5x = 768$
 $\Rightarrow 5x = 615 \Rightarrow x = 123$

13. (b) Let H.C.F. be x
 According to question,
 $x \times \text{L.C.M} (6, 7, 3) = 3654$
 $\Rightarrow x = \frac{3654}{6 \times 7} = 87$

14. (a) According to question,
 S.I. $= \frac{3000 \times 7 \times 5}{100} = ₹1050$

15. (c) Let radius of big hemisphere $= r$
 According to question,
 $\frac{80}{100} \times 20 \times \frac{2}{3} \times \pi \times (3)^3 = \frac{2}{3} \pi r^3$
 $\Rightarrow 8 \times 2 \times (3)^3 = r^3 \Rightarrow r = 6\sqrt{2} \text{ cm}$

16. (b) Average production of Company X
 $= \frac{8.1 + 8 + 5.2}{3} = \frac{21.3}{3} = 7.1$

Average production of Company Y

$$= \frac{7.6 + 6.6 + 6.5}{3} = \frac{20.7}{3} = 6.9$$

Average production of Company Z

$$= \frac{7 + 7.2 + 6.2}{3} = \frac{20.4}{3} = 6.8$$

Average production of Company W

$$= \frac{7 + 6.9 + 6.8}{3} = \frac{20.7}{3} = 6.9$$

$\therefore$ Maximum average $= X$

17. (a) Successive discount $= 12 + 6 - \frac{12 \times 6}{100}$
 $= 17.28\%$
 Required amount
 $= 2400 \times \frac{82.72}{100} = ₹1985.28$

18. (a) 20% of number of boys

$$= \frac{3}{4} \times \text{number of girls}$$

$$\Rightarrow \frac{1}{5} \times \text{boys} = \frac{3}{4} \times \text{girls}$$

$$\Rightarrow \frac{\text{boys}}{\text{girls}} = \frac{15}{4}$$

19. (c) When a person sells two similar items, one at a gain of say $y\%$ and other at a loss of $y\%$ then the seller always incurs a Loss.

$$\therefore \text{Loss}\% = \frac{y^2}{100}\%$$

$$\therefore \text{Loss}\% = \frac{(20)^2}{100} = 4\%$$

20. (a) Total number of technicians who left the company = $50 + 160 + 200 + 180 = 590$
Total number of technicians who joined the company = $260 + 340 + 450 + 480 = 1530$
Required % = $\frac{590}{1530} \times 100 = 38.56\% \approx 39\%$

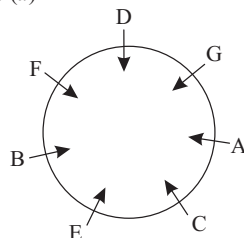
21. (c) $39 \div 3 - 24 + 60 \times 12$

After interchanging the signs,
 $= 39 \times 3 + 24 - 60 \div 12$
 $= 117 + 24 - 5 = 136$

22. (b) (I) swim fast $\rightarrow$ cp (lk) bs

may (I) go $\rightarrow$ ap sb (lk)
 By Comparing both the codes,
 Code of 'I' is 'lk'.

23. (a)



Two people sitting between D and E.

24. (b)



25. (a) W I (N) G $\rightarrow$ 2 (6) 1 8

(N) E A T $\rightarrow$ 3 4 (6) 9
 Code of 'N' is '6'.

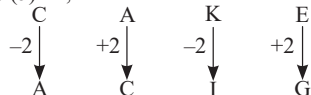
26. (a) Logic: Second Number = First number + third number + 20

As, $(4, 35, 11) \Rightarrow 4 + 11 + 20 = 35$

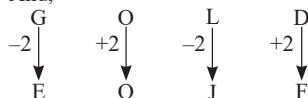
And, $(1, 82, 61) \Rightarrow 1 + 61 + 20 = 82$

Similarly, $(13, 53, 20) \Rightarrow 13 + 20 + 20 = 53$

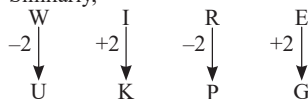
27. (b) As,



And,



Similarly,



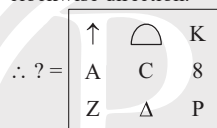
28. (c)

1	2	3
4	5	6
7	8	9

Logic:

I. For getting even place figure, Letters or symbols in place 1, 2, 3, 6, 9 rotates one steps in anticlockwise direction and letters or symbols in 4, 5, 8, 7 rotates one steps in clockwise direction.

II. For getting odd place figure, letters or symbols in place 1, 9, 8, 7, 4 rotates one steps in clockwise direction and letters or symbols in 2, 3, 6, 5 rotates one steps in clockwise direction.

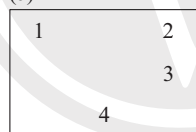


$\therefore ? =$

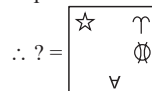
29. (c) Chicken hens $\times$ egg

Neither conclusion I nor II follows.

30. (c)

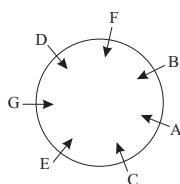


Logic: Symbols in place 1, 2, 3 rotates one steps in clockwise direction.



$\therefore ? =$

31. (b)



G sits third to the left of A.

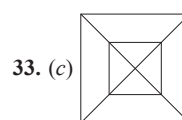
32. (c) Logic: First number $\times 10 - 40$

= second number

As, $(7, 30) \Rightarrow 7 \times 10 - 40 = 30$

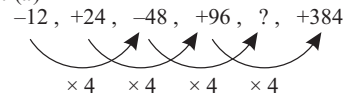
And, $(14, 100) \Rightarrow 14 \times 10 - 40 = 100$

Similarly, $(22, ?) \Rightarrow 22 \times 10 - 40 = 180$



33. (c)

34. (a)

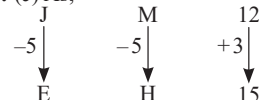


$$\therefore ? = -48 \times 4 = -192$$

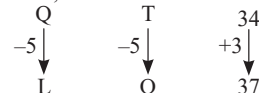
35. (d) $5 \div 6 \times 3 - 4 + 11 = ?$

After interchanging the signs,
 $= 5 \times 6 \div 3 + 4 - 11 = 5 \times 2 + 4 - 11$
 $= 10 + 4 - 11 = 3$

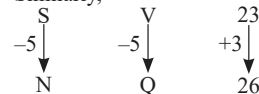
36. (c) As,



And,



Similarly,



37. (d)



Both conclusions I and II follow.

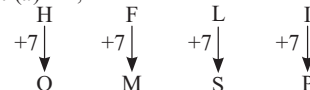
38. (a) Logic: Number of letters in word $\times 8$

As, COLONY = $6 \times 8 = 48$

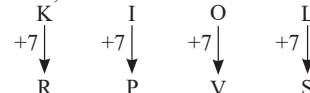
And, INTERCHANGE = $11 \times 8 = 88$

Similarly, MOUNTAIN = $8 \times 8 = 64$

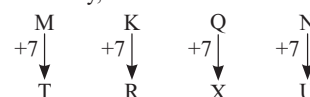
39. (d) As,



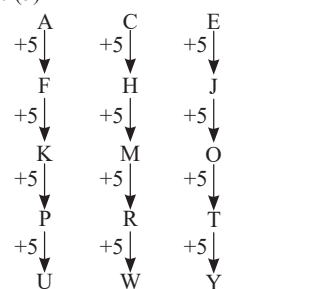
And,



Similarly,



40. (b)



41. (d) The Pharmacy (Amendment) Bill, 2023 was introduced in the Lok Sabha on August 3, 2023, which amends the Pharmacy Act, 1948.
The Pharmacy (Amendment) Bill, 2023 seeks to revise the Pharmacy Act, 1948, which oversees the pharmacy profession in India.
The Act's goal is to make sure that only trained and qualified people work as pharmacists, ensuring the safety of public health.
42. (c) Losar is celebrated in the Union Territory of Ladakh and the state of Arunachal Pradesh to mark the beginning of the new year.
"Losar" is the most festive celebration all year around.
It is celebrated to mark the beginning of the Ladakhi New Year.
The festival is celebrated on the 1st day of the eleventh month of Tibetan Calendar.
43. (a) Kanak Rele was an Indian dancer, choreographer, and scholar, widely recognized as a leading figure in Mohiniyattam.
She established the Nalanda Dance Research Centre and served as the founding principal of Nalanda Nritya Kala Mahavidyalaya in Mumbai.
Mohiniyattam is an Indian classical dance form that developed and remained popular in the state of Kerala.
44. (b) The system of Diarchy, meaning dual government, was introduced in British India through the Montagu-Chelmsford Reforms and the Government of India Act, 1919.
In this system, provincial administration was divided into two parts: 'transferred' subjects, handled by elected ministers, and 'reserved' subjects, controlled by the Governor and his officials.
45. (c) India's northern region, especially the Himalayan belt, experiences a cool mountain (alpine) climate.
The high altitude areas of states like Himachal Pradesh, Uttarakhand, and parts of Jammu & Kashmir, and Ladakh have cooler temperatures and receive snowfall in winter.
These regions are significantly cooler compared to other parts of India due to their elevation and mountainous terrain.
46. (c) Chandragupta Maurya established the Mauryan Empire in 322 BCE after overthrowing the Nanda dynasty.
The empire's capital was at Pataliputra (near modern-day Patna), which became a significant political and cultural centre.
Under his leadership, the Mauryan Empire became one of the most expansive and consolidated empires in ancient India.
47. (c) Shafiqur Rahman Barq is a veteran politician associated with the Samajwadi Party (SP).
He was elected to the Lok Sabha at various times from Moradabad as the Samajwadi Party candidate and Sambhal, most recently in 2019.
48. (c) Dadabhai Naoroji, a prominent nationalist leader, educator, and intellectual, wrote "Poverty and Un-British Rule in India." He was one of the founding members of the Indian National Congress and served as its 2nd, 9th, and 22nd President from 1886 to 1887, 1893 to 1894 and 1906 to 1907.
India Post depicted Naoroji on stamps in 1963, 1997 and 2017.
49. (d) K.M. Munshi, a noted Indian writer and historian, primarily wrote his literary works 'The Glory of Patan' in Gujarati.
'The Glory of Patan' was originally published as 'Patan ni Prabhuta' in Gujarati, reflecting the rich cultural and historical heritage of medieval Gujarat.
K.M. Munshi was popularly known by his pen name 'Ghanshyam Vyas'.
50. (d) The Punjab Legislative Council (Abolition) Act, 1969 was a legislation in India, adopted in 1969. Punjab, along with some other states in India, had a bicameral legislature that included a Legislative Assembly and a Legislative Council.
Punjab abolished its Legislative Council on 1 January 1970 to simplify its legislative structure and reduce administrative costs.
51. (a) Turmeric, a widely used spice in Indian and Southeast Asian cuisine owes its bright yellow colour and many of its health benefits due to the compound curcumin.
Curcumin is a bioactive substance that has antioxidant and anti-inflammatory properties.
Unlike tartaric acid (commonly found in fruits like grapes), carotene (associated with the orange colour of carrots), or capsaicin (the heat-inducing compound in chilli peppers), curcumin is unique to turmeric.
52. (b) The Indore gharana, a renowned style of Hindustani classical music, was established by Ustad Amir Khan (1912–1974).
Bhimsen Joshi was associated with the Kirana Gharana.
Tansen was a legendary musician in the Mughal court but did not establish the Indore Gharana.
Ustad Ghaghe Khuda Baksh is not recognized as the founder of any major gharana.
53. (d) Pietra Dura, an Italian term, refers to the technique of using inlaid precious and semi-precious stones to create intricate floral and geometric patterns in architecture.
Pietra dura items are generally crafted on green, white or black marble base stones.
- This technique found extraordinary expression in Mughal monuments, most notably in the Taj Mahal in Agra.
54. (b) In India, the agricultural year is divided broadly into two main cropping seasons: Rabi and Kharif.
The Rabi cropping season in India starts with the sowing of seeds in October–November, coinciding with the onset of the winter season.
It concludes with the harvest in March–April, just before the arrival of the summer months.
55. (c) Sodium bicarbonate, commonly known as baking soda, has the chemical formula NaHCO_3 . It is a mild base used in cooking, cleaning, and various chemical reactions.
Sodium bicarbonate releases carbon dioxide when it reacts with acid or is heated, making it a key ingredient in baking to help dough rise.
 NaCl is sodium chloride (common salt), Na_2CO_3 is sodium carbonate (washing soda), and NaOH is sodium hydroxide (caustic soda).
56. (a) Capital goods are assets used by businesses to produce other goods or services. They are typically long-lasting (durable) and have a prolonged life cycle. Examples include machinery, tools, and equipment.
They are neither "Giffen goods" (a rare category of goods where demand increases as price rises due to lack of cheaper substitutes) nor "inferior goods" (goods for which demand decreases as consumer income rises).
57. (c) Based on their prestige, the correct decreasing order of the Padma awards is Padma Vibhushan followed by Padma Bhushan and then Padma Shri.
Padma Vibhushan: This is the second-highest civilian award, given for exceptional and distinguished service without distinction of category.
Padma Bhushan: This is the third-highest civilian award, recognizing exceptional and distinguished service of a high order.
Padma Shri: This is the fourth highest civilian award in India. It is awarded for distinguished contributions in various fields such as arts, education, industry, literature, science, and public service.
58. (d) The Vijayanagara Empire was founded in the early 14th century by Harihara I and Bukka Raya I, who belonged to the Sangama dynasty.
They established the kingdom along the southern banks of the Tungabhadra River in present-day Karnataka.
The later ruling dynasties such as Saluva, Tuluva, and Aravidu came into power after the Sangama lineage.

59. (b) Tomatoes contain two pigments for photosynthesis—chlorophyll, which is green, and lycopene, which is red. When tomatoes start to grow, they contain much less lycopene than chlorophyll, which gives them their green color. But when harvest season arrives, the days shorten and temperatures drop, causing chlorophyll to dissolve and lycopene to take over the shade of the fruit.
60. (b) According to the 2011 census, the lowest recorded literacy rates are in the Dadra and Nagar Haveli. In Dadra and Nagar Haveli, the overall literacy rate is 77.24 percent. It has the lowest male literacy rate in India. The literacy rate for men is 85.17%, while the literacy rate for women is 64.32%.
61. (c) Fundamental Duties were introduced into the Indian Constitution by the 42nd Amendment Act, 1976, under Part IV A (Article 51A) of the Constitution. The Swaran Singh Committee recommended Fundamental Duties in 1976, the necessity of which was felt during the internal emergency of 1975-77. The 42nd Amendment Act of 1976 added 10 Fundamental Duties to the Indian Constitution. The 86th Amendment Act 2002 later added the 11th Fundamental Duty to the list.
62. (a) Sanjana Bathula is an Indian athlete associated with roller skating. She was the reserve member for the Indian team which won a bronze medal in the women's 3000m relay event at the 2022 Asian Games, but she didn't receive a medal.
63. (d) Smriti Mandhana, a top Indian batter, was the captain of the Royal Challengers Bangalore (RCB) women's team in the first Women's Premier League (WPL) in 2023. Harmanpreet Kaur captained the Mumbai Indians, Deepti Sharma played for UP Warriorz, and Shafali Verma was a dynamic batter for Delhi Capitals. Royal Challengers Bangalore won the Women's Premier League 2024 (Cricket) in India against Delhi Capitals on 17 March 2024.
64. (d) Mawsynram is the wettest place on Earth due to its extremely high annual rainfall. It is located in the Meghalaya state of India. Hence, both statements A and B are correct. The region's unique topography and climatic conditions cause moisture-laden winds from the Bay of Bengal to produce intense and frequent rainfall.
65. (b) India shares its land borders with seven countries. These are Pakistan to the west, China, Nepal, and Bhutan to the north, and Bangladesh and Myanmar to the east. Afghanistan also shares a small border segment with India in the region of Gilgit-Baltistan (now occupied by Pakistan), but it is not recognized as a current practical border in the contemporary geopolitical scenario post-partition.
66. (c) Consicstent – Incorrect spelling. The correct spelling is Consistent.
67. (b) Happy (खुश): Feeling or showing pleasure or contentment. Ecstatic (उत्साही): Extremely happy or joyful. Shocked (हैरान): Surprised greatly, often with an emotional impact. Angry (गुस्से में): Feeling or showing strong displeasure or hostility. Interested (रुचि रखने वाला): Showing curiosity or concern about something.
68. (c) "Hand down" means to give or pass on something, usually from one person to another. The correct phrasal verb here should be "hand over," as it refers to physically passing something e.g., payment slips.
69. (b) "Reading between the lines" refers to understanding a deeper or hidden meaning beyond the literal interpretation.
70. (c) The workers had been repairing Past perfect continuous is incorrect; it should be "have been repairing".
71. (b) The idiom "a piece of cake" means something is very easy to do.
72. (c) "Civilisation" is the most appropriate term. Cultural heritage is indeed an essential part of any civilisation, and it connects people across generations.
73. (c) "Rides" is the simple present form, which is the appropriate tense for habitual actions or routines as in "every day".
74. (a) Chase (पीछा करना): To follow or seek after something with determination. Pursue (अनुसरण करना): To engage in or continue with an activity, goal, or ambition, often with effort. Inspired (प्रेरित): To fill someone with the urge or ability to do something. Immense (विशाल): Extremely large or great. Success (सफलता): The accomplishment of an aim or purpose.
75. (a) Should be "is disposed to" instead of "disposed of" to indicate a tendency or inclination.
76. (c) accident is Correct spelling.
77. (b) Should be "was in" as "Neither" is singular.
78. (c) Challenging (चुनौतीपूर्ण): Requiring great effort, skill, or determination. Gruelling (कठिन): Extremely tiring or demanding; physically or mentally exhausting. Empowering (सशक्त बनाना): Making someone stronger or more confident. Inclusive (समावेशी): Including all people or things. Accessible (सुलभ): Easy to approach or reach.
79. (b) Meaning "excessively tired".
80. (d) Correct adverb for listening.
81. (d) Power (शक्ति): The ability or capacity to do something. Energy (ऊर्जा): The capacity to do work or produce change; physical or mental strength. Frailty (दुर्बलता): The condition of being weak or fragile. Strife (संघर्ष): Conflict or struggle. Speed (गति): The rate at which something moves or operates.
82. (c) Perceive (समझना): To become aware of or understand something through the senses or the mind. Discern (पहचानना): To recognize or identify something clearly, especially after careful observation or analysis. Ignore (नजर अंदाज करना): To deliberately pay no attention to something. Confuse (भ्रमित करना): To cause someone to become unclear or uncertain about something. Fabricate (बनाना): To invent or make up something, usually with the intention to deceive.
83. (c) Incorrect subject-verb agreement, should be "watch".
84. (d) hurricane is Correct spelling.
85. (a) Meaning understanding or realization.
86. (a) A pigeon refusing to "fly" makes sense, as it is a bird that typically flies, but in this case, it is refusing to do so.
87. (b) "On" is the correct preposition when saying "come on walks," which is the idiomatic usage for joining someone on a walk.
88. (b) "Either" is used in "either...or" constructions to indicate one of two options, which fits here with "you carried him on your shoulder or let him walk."
89. (c) "Pace" refers to the speed or rate of movement, which fits the context of adjusting walking speed to match the pigeon's pace.
90. (c) "Indignation" fits better, as it refers to the pigeon's distress or frustration while running after you.